14.5.1 Need for Constructors

An array or a structure in c++ can be initialized during the time of their declaration.

For example

```
struct sum
  {
        int n1,n2;
  };
class add
  {
        int num1,num2;
  };
int  main()
  {
        int arr[]={1,2,3};        //declaration and initialization of array
        sum s1={1,1};            //declaration and initialization of structure object
        add  a1={0,0};          // class object declaration and initialization throws
                                                          compilation error
  }
```

Member function of a class can access all the members irrespective of their associated access specifier.

The initialization of class type object at the time of declaration similar to a structure or an array is not possible because the class members have their associated access specifiers (private or protected or public). Therefore Classes include special member functions called as *constructors*. The constructor function initializes the class object.

14.6 Declaration and Definition

When an instance of a class comes into scope, a special function called the *constructor* gets executed. The constructor function name has the same name as the class name. The constructors return nothing. They are not associated with any data type. It can be defined either inside class definition or outside the class definition.

Example 1:

Illustration 14.5 A constructor defined inside the class specification.

```
#include<iostream>
using namespace std;
class Sample
{
        int i,j;
        public :
          int k;
          Sample()
          {
             i=j=k=0;   //constructor defined inside the class
          }
};
```


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